generalManagement

Service Description

Service ID: *"generalManagement"*

Abstract
This document describes the `generalManagement` service, version 1.0.0, produced by the DeviceQoSEvaluator support system of Arrowhead 5.2.2.

Contents

1 Overview **3**
1.1 Significant Prior Art . . . 3
1.2 How This Service Is Meant to Be Used . . . 3
1.3 Important Delimitations and Dependencies . . . 3

2 Service Interface **4**
2.1 operation get-log . . . 4
2.2 operation get-config . . . 4

3 Information Model **5**
3.1 struct KeyValues . . . 5
3.2 struct LogEntry . . . 5
3.3 struct LogEntryListResponse . . . 5
3.4 struct LogRequest . . . 5
3.5 struct Page . . . 5
3.6 Primitives . . . 6

4 References **7**

5 Revision History **8**
5.1 Amendments . . . 8
5.2 Quality Assurance . . . 8

1 Overview

This document describes the `generalManagement` service, which provides general management of the Device QoSEvaluator system: reading its log entries and its configuration. It is produced by the DeviceQoSEvaluator system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

1.1 Significant Prior Art

New in Arrowhead 5; every 5.x core system offers it.

1.2 How This Service Is Meant to Be Used

Operators and LocalCloudManager read recent log entries (stored in the system's database) with `get-log` and the effective configuration with `get-config`.

1.3 Important Delimitations and Dependencies

Access is governed by the management policy of DeviceQoSEvaluator. `get-config` returns only non-secret configuration keys.

2 Service Interface

This section describes the interfaces to the `generalManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

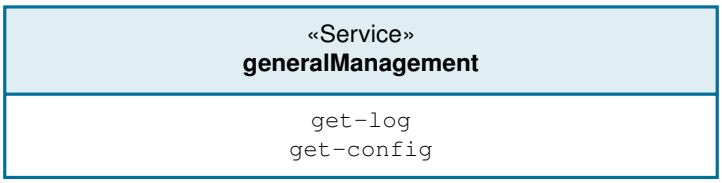


Figure 1: SysML-style block of the `generalManagement` service and its operations.

The following interface operations are available.

2.1 operation get-log (LogRequest) : LogEntryListResponse

Returns a page of the system's log entries matching the filter.

2.2 operation get-config () : KeyValues

Returns the effective values of the requested configuration keys. Only the public keys of the system are answered; other keys, including all secrets, are ignored.

3 Information Model

Here, all data objects that can be part of the `generalManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.6, which are used to represent things like hashes and identifiers.

3.1 struct KeyValues

Configuration values of the system.

Field	Type	Description
map	Map<String, String>	The configuration keys and values; secrets are not returned.

3.2 struct LogEntry

One log entry.

Field	Type	Description
logId	Identifier	Identifier of the log entry.
entryDate	DateTime	UTC time of the log entry.
logger	String	Logger (Java class) that wrote the entry; as filter, its name.
severity	String	Log level, e.g. `INFO`, `WARN`, `ERROR`; as filter, the minimum level.
message	String	Text of the log entry or status message of the job.
exception	String	Stack trace or exception text of the log entry, if any.

3.3 struct LogEntryListResponse

List of log entries.

Field	Type	Description
entries	List<LogEntry>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.4 struct LogRequest

Paged filter for the log entries stored in the system's database.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
from	DateTime	Return entries from this UTC time.
to	DateTime	Return entries until this UTC time.
severity	String	Minimum log level to return.
logger	String	Logger (Java class) that wrote the entry; as filter, its name.

3.5 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

3.6 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Number	An integer number.
String	A chain of characters.

4 References

[1] J. Delsing, “DeviceQoSEvaluator – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/DeviceQoSEvaluator/documentation/v5.2.2/DeviceQoSEvaluator-v5.2.2-SysD.pdf.

[2] ——, “generalManagement HTTP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-generalManagement-HTTP-IDD.pdf.

[3] ——, “generalManagement MQTT/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-generalManagement-MQTT-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	